

Near-Field Photometric Stereo for CT-Registered Surface Reconstruction with a Near-Coaxial Monocular Endoscope: Millimetre Accuracy on Spinal Bone

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Purpose. Image-guided spine surgery depends on registering the exposed bony surface to a preoperative CT. We test whether a monocular endoscope with a ring of near-coaxial LEDs — the geometry forced by an 18 mm scope — can reconstruct the exposed spinal surface accurately enough for per-vertebra CT registration, using only its own images and no added tracking hardware.

Methods. Four single-LED frames per pose drive a photometric-stereo solve. We compare far-field photometric stereo, a near-field point-source model (per-pixel light directions + inverse-square attenuation), and an inverse-square brightness-to-range baseline. Each reconstruction is anchored to CT through retroreflective fiducials visible in both modalities, via per-vertebra Perspective- n -Point pose and an affine depth calibration. Accuracy is the 3-D fiducial registration error (FRE) against CT on five vertebral levels.

Results. With the near-field model, three of five levels reconstructed to **1.2–1.7 mm** 3-D FRE with four fiducials each — comparable to clinical surface-based spine registration (≈ 1 mm, 20 points) and better than reported in-vivo endoscopic photometric stereo (< 3 mm). Near-field modelling removed the systematic “bowl” of far-field integration (worst far-field level improved from 6.6 to 1.7 mm FRE). The inverse-square baseline failed (7.8–9.6 mm, physically inverted depth). All levels were imaged at a similar working distance (51–60 mm); the two non-millimetre levels failed for non-distance reasons (three fiducials; an unresolved near-field divergence), reported in full.

Conclusion. Near-coaxial monocular photometric stereo yields per-vertebra, CT-registered surfaces at millimetre accuracy on well-conditioned levels, without added radiation or tracking hardware; the dominant residual is the use of assumed rather than calibrated LED positions.

Keywords photometric stereo · near-field · endoscopy · image-guided surgery · spine · CT registration · fiducial registration error

1 Introduction

Spinal navigation registers each vertebra as an independent rigid body — the spine flexes at the intervertebral discs, so no single rigid transform fits multiple levels — using intraoperative imaging (cone-beam CT / O-arm, adding radiation) or a tracked surface digitiser. An optical method that reconstructs the exposed surface from the endoscope already in the field, with no added radiation and no external tracker, is attractive.

Metric shape from a single moving endoscope is hard: structure-from-motion and SLAM need texture and parallax that wet bone provides poorly, and neural rendering needs many views and does not natively produce a CT-registered metric surface. *Photometric stereo* is attractive because it is *albedo-invariant*: it recovers the surface normal from *ratios* of brightness across illuminations, so a dark and a bright patch at the same tilt give the same normal — exactly the property brightness-based shape-from-shading lacks on textured bone.

The obstacle is geometry. A ring of LEDs around an 18 mm scope sits ≈ 6 mm off the optical axis, so at 40 mm working distance the lights are *near-coaxial* ($\approx 8.6^\circ$ off-axis; $\approx 17^\circ$ between opposing pairs) and *near-field* (direction and intensity vary across the field). Far-field photometric stereo mis-models this and integrates the residual brightness gradient into a spurious low-frequency curvature — the “bowl”.

Contributions. (1) A near-field photometric-stereo pipeline for a near-coaxial LED endoscope that removes the far-field bowl; (2) a per-vertebra CT registration protocol using dual-modality fiducials, PnP pose, and affine depth calibration, validated by 3-D FRE; (3) a controlled far-field / near-field / inverse-square

comparison on CT-validated data; (4) a transparent failure analysis (three fiducials under-constrain PnP; an unresolved near-field divergence) and isolation of assumed LED positions as the dominant residual error.

2 Related work

Woodham's far-field photometric stereo [1] recovers normals from ≥ 3 directional illuminations; Frankot–Chellappa integration [2] yields depth. Quéau et al. [3] give a complete LED forward model with calibration; robust point-light *position* calibration reaches < 2.7 mm [4], and multi-view near-field benchmarks now exist [5]. In-vivo endoscopic photometric stereo has reconstructed the colon at < 3 mm mean depth error [6]; neural and Gaussian-splatting pipelines [7] excel at rendering but do not deliver a CT-registered metric surface. Surface-based spine registration reaches ≈ 0.96 mm RMS with ~ 20 lamina points [8]; quality is reported as FRE/TRE [9]. We target metric, CT-validated, *per-vertebra* registration on an unusually near-coaxial rig.

3 Materials and methods

3.1 Imaging system

A monocular endoscope (native 640×480) carries four LEDs on an ≈ 6.05 mm ring. At 40 mm each LED is $\approx 8.6^\circ$ off-axis; opposing LEDs subtend $\approx 17^\circ$, so the tilt-encoding lateral component is only $\approx \cos 81^\circ \approx 15\%$ of each light vector. Field of view $\approx 37\text{--}43$ mm. Retroreflective fiducials appear in CT as small surface craters, giving dual-modality correspondences. Each capture records a dark frame and four single-LED frames.

3.2 Calibration & preprocessing

Intrinsics from a ChArUco target; working focal length ($f_x \approx 860$ px) validated by 4-point PnP reprojection (0.42 px residual). Frames are dark-subtracted, gamma-

linearised, specular-inpainted, and per-frame median exposure-balanced.

3.3 LED index → azimuth mapping

The physical wiring is LED1→top (90°), LED2→bottom (270°), LED3→left (180°), LED4→right (0°). An incorrect assumption produced normals *anti-correlated* with CT ($r \approx -0.73$); the correct mapping gives $r \approx +0.97/+0.99$ on two CT-validated levels — a common silent failure mode for multi-LED rigs.

3.4 Photometric-stereo models

Far-field: constant per-LED direction, solve $I_k = \rho(\mathbf{n} \cdot \mathbf{L}_k)$ and integrate. *Near-field*: for a back-projected point at depth D and LED position $\mathbf{P}_s$, the incident direction and $1/r^2$ attenuation are computed per pixel; writing $m_s = I_s \cdot \|\mathbf{P}_s - \mathbf{X}\|^2$ linearises the point-source model to a per-pixel least-squares solve, iterated with a CT-anchored depth (4 iterations). LED positions are *assumed* from the ring geometry — the principal approximation.

3.5 Per-vertebra CT registration

Four non-coplanar fiducials give the CT→camera pose (SQPnP). Photometric depth is affine-calibrated, $z_{\text{lens}} = a \cdot z_{\text{photo}} + b$, on the fiducials; the surface is back-projected and transformed into CT coordinates. Each vertebra is registered independently (the spine flexes); a single rigid multi-level fit is used only as a rigidity check. The dense surface is masked (brightness, photometric-residual confidence, largest connected component, border crop) and mask-aware smoothed before meshing.

4 Results

Table 1 — 3-D fiducial registration error (FRE), far-field vs near-field. All levels imaged at a similar working distance (51–60 mm); levels 4–6 are the well-conditioned 4-marker cases.

Level	WD (mm)	Far-field FRE	Near-field FRE	Note
4	60.0	1.54 mm	1.72 mm	

Level	WD (mm)	Far-field FRE	Near-field FRE	Note
5	55.4	1.22 mm	1.32 mm	
6	58.9	6.57 mm	1.74 mm	bowl removed
7	58.6	11.14 mm	9.72 mm	3 markers
8	51.5	5.66 mm	diverged	near-field

4.1 Far-field vs near-field

On levels 4–6 the near-field model gives **1.2–1.7 mm** FRE with four fiducials each. The decisive gain is level 6 (6.6→1.7 mm). Fig. 1 shows the same shot, same registration: the far-field surface bows while the near-field surface flattens. Even on level 4 — registered to 1.5 mm — the far-field surface arches between fiducials; near-field reduced the plane-fit deviation from 6.90 to 4.98 mm (part of the residual is genuine bone relief). Figs. 2–3 show near-field surfaces registered to CT with reconstructed fiducials on the ground truth.

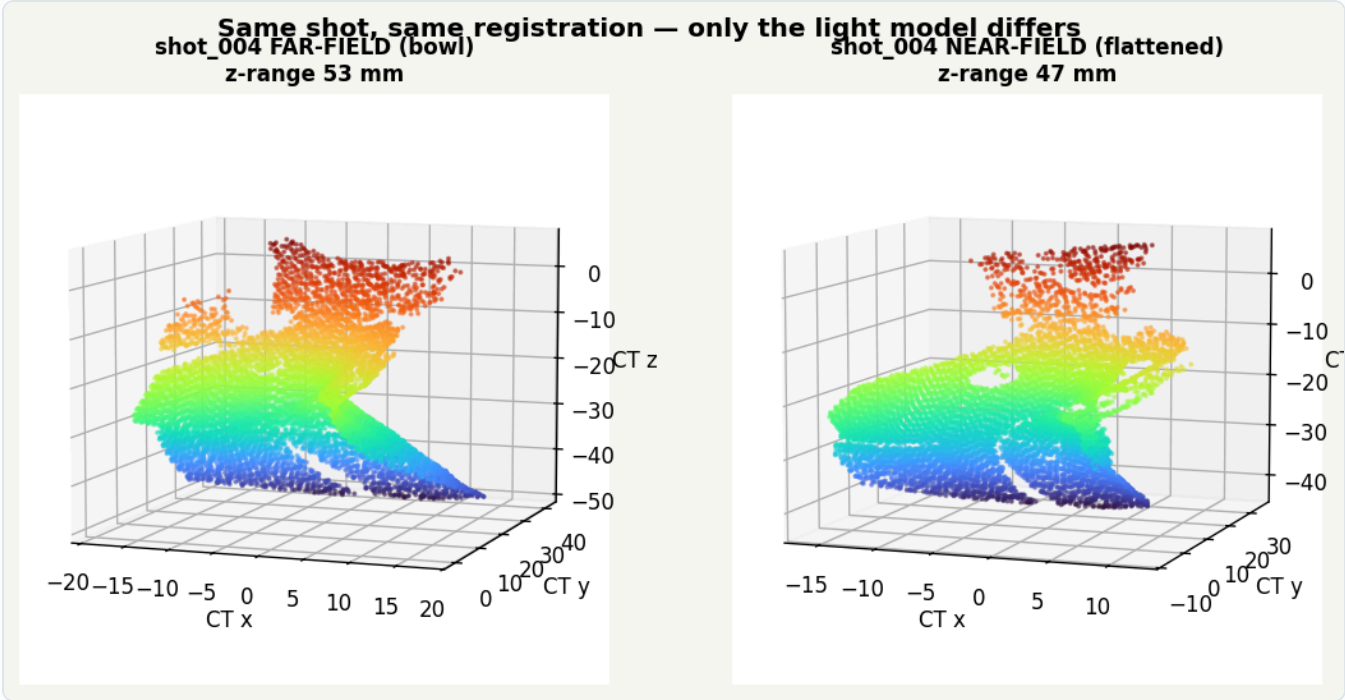


Figure 1. Level 4, identical registration — only the light model differs. Far-field (left) integrates a bowl; near-field (right) flattens it.

shot_006 — NEAR-FIELD surface registered to CT
3-D marker RMS 1.7 mm, relief 29.4 mm

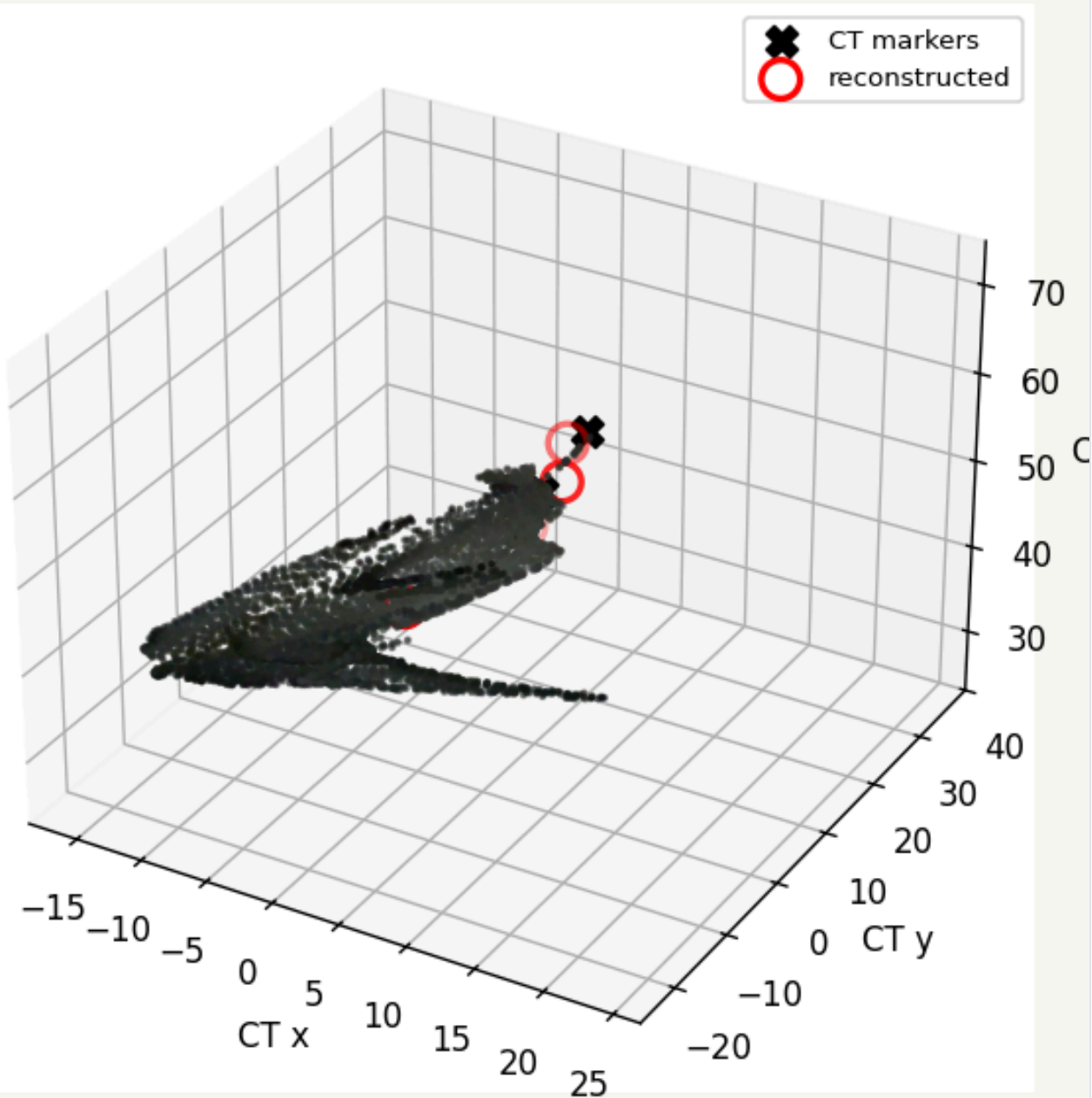


Figure 2. Level 6 near-field surface registered into CT coordinates; reconstructed fiducials (open circles) on CT ground truth (crosses). 3-D FRE 1.7 mm, versus 6.6 mm far-field.

shot_004 — NEAR-FIELD surface registered to CT
3-D marker RMS 1.7 mm, relief 18.7 mm

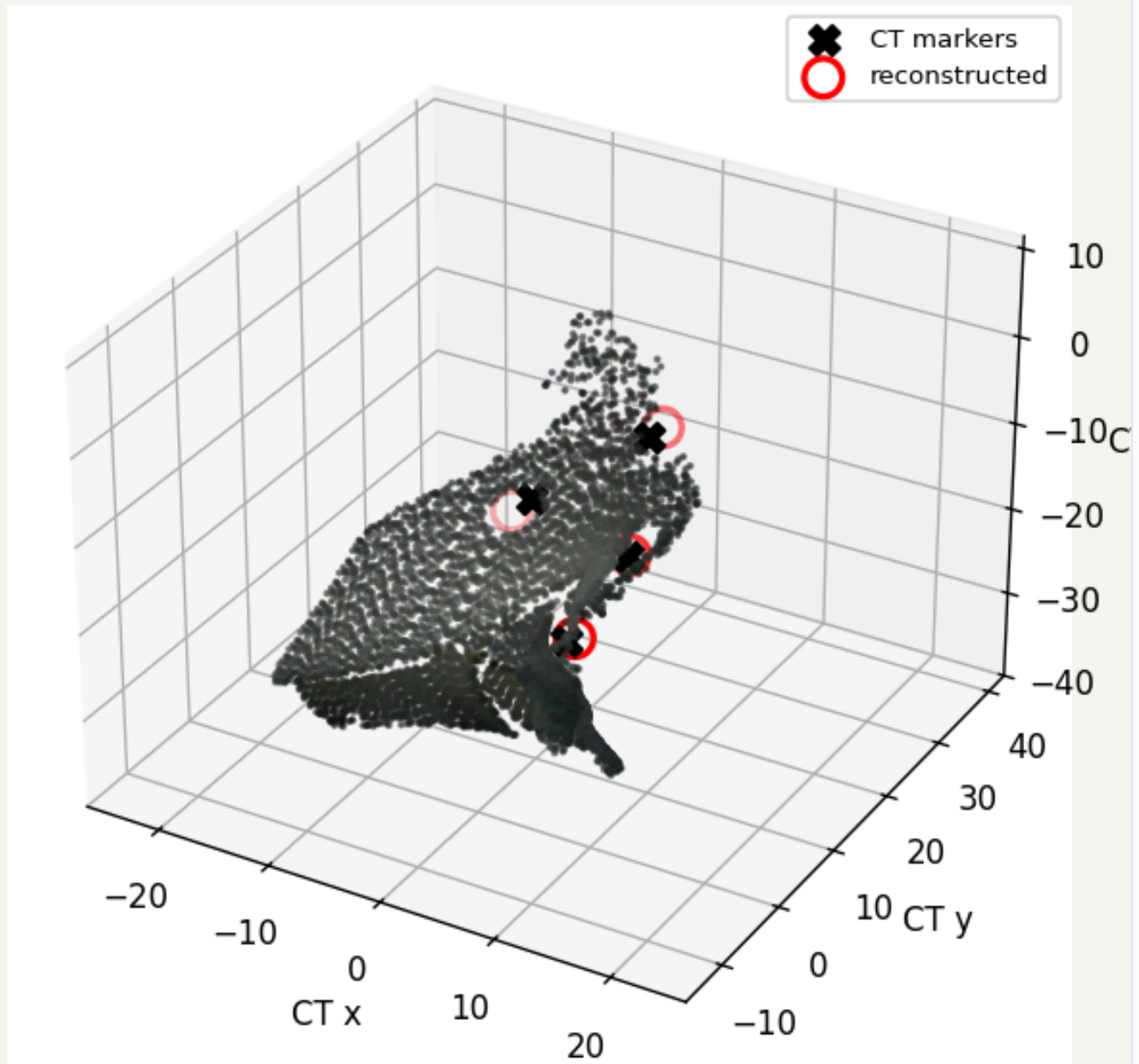


Figure 3. Level 4 near-field surface registered to CT; 3-D FRE 1.7 mm.

4.2 Inverse-square baseline

A brightness-to-range baseline ($r \propto 1/\sqrt{I}$) with identical CT anchoring failed: 7.8–9.6 mm FRE, dense relief inflating to 150–559 mm (true CT range 17–26 mm), and a physically inverted depth slope. The cause is intrinsic — inverse-square assumes constant albedo, so dark bone reads as “far” — quantifying why the normal-based solve is necessary.

4.3 Failure modes

All five levels were imaged at a similar working distance (51–60 mm), so the two failures are *not* distance-driven. Level 7 has only three fiducials, leaving the PnP pose under-constrained (P3P admits up to four solutions). Level 8 has the *best* marker geometry, yet the near-field iteration diverged and its confidence mask collapsed, while the far-field solve on the same data stayed stable (5.66 mm) — an unresolved solver-robustness issue, reported rather than omitted. Because working distance was effectively fixed here, its effect on accuracy could not be characterised; a controlled distance sweep is future work.

5 Discussion

Table 2 — Accuracy in context.

Method / study	Accuracy	Notes
This work (near-field, 4–6)	1.2–1.7 mm	4 fiducials/level
Surface-based spine reg. [8]	~0.96 mm	20 points, clinical
In-vivo endoscopic PS [6]	<3 mm	colon, depth error
Near-field light calib. [4]	<2.7 mm	light position

Our millimetre accuracy sits between clinical surface registration and reported in-vivo endoscopic photometric stereo, with fewer fiducials and a harder near-coaxial geometry, and — unlike appearance-oriented neural methods — yields a directly CT-registered metric surface. The contribution is that on a near-coaxial endoscopic rig the far-field approximation injects a systematic shape error a global scale/offset cannot remove, and a per-pixel point-source model removes it while preserving albedo-invariance.

Limitations. (1) *Assumed LED positions* bound the residual bowl; a mirror-sphere / matte-target calibration [3,4] is the next step, expected to move levels 6–8 toward 1 mm. (2) Near-coaxial low SNR compresses recovered relief. (3) Reliable only to ~55 mm. (4) Three fiducials under-constrain pose (level 7); specular markers are

inpainted, leaving small holes. (5) Bench validation on one instrumented specimen; wet in-vivo fields are future work.

Clinical relevance. Per-vertebra registration matches spine navigation practice; the combined multi-level result is a bench consistency check. A calibrated CT-registered optical surface from the existing scope could reduce reliance on intraoperative radiation for re-registration.

6 Conclusion

A near-coaxial monocular endoscope, driven by four-LED near-field photometric stereo and anchored to CT through dual-modality fiducials, reconstructs the exposed spinal surface to **1.2–1.7 mm** per vertebra on well-conditioned levels — competitive with clinical surface registration and better than reported in-vivo endoscopic photometric stereo, without added radiation or tracking hardware. The far-field bowl is removed; the inverse-square shortcut is shown inapplicable to textured bone; and the dominant residual is localised to assumed LED positions, defining the next experiment.

References

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A Publication strategy

Primary target — IJCARS (Int. J. Computer Assisted Radiology & Surgery), IF ≈ 2.8 , Q1. Scope is exactly computer-assisted interventions, navigation, and registration. *Recommended first submission.*

Alternatives: IEEE TBME (IF ≈ 4.4 , Q1 — lead with the near-field modelling); Biomedical Optics Express (Q1 — lead with the optical / LED novelty); Medical Physics (Q1 — rigorous validation home).

Reach (after LED calibration + a second specimen): Medical Image Analysis and IEEE TMI (IF ≈ 10 , top Q1).

What Q1 reviewers will expect: CT ground-truth metric (have it: 3-D FRE); a model ablation (have it: far / near / inverse-square); disclosed limits (have them: the two failure modes); reproducible code (release `marker_pipeline/`); and — before a top-tier attempt — a *calibrated* LED forward model plus ≥ 2 specimens to convert “millimetre on a bench” into a clinical accuracy claim.